Assessment of endogenous androgen levels in meat, liver and testis of Iranian native cross-breed male sheep and bull by gas chromatography-mass spe...

Androgenic steroids always exist in different animal tissues at trace level, with significant numbers of interfering compounds, which makes their determination difficult. To solve some of the problems in quantification of the natural steroids in those tissues, a new GC-MS method was developed in this study. By using a surrogate analyte approach, which was developed in the authors' previous studies, and extensive sample preparation procedure, which successfully eliminates many of the interfering compounds and resulting in a cleaner extract, accuracy, precision, sensitivity and selectivity of the method for the determination of steroids in complex matrices such as meat, liver and testis were improved. By aid of this method, the levels of androgens in different tissues of Iranian native cross-breed bulls and male sheep were determined. According to the results obtained in the present study, although the androgenic profile (contents and ratios of precursors and metabolites to the main hormones) is similar between the same tissues of both animals, the total androgenic content of each tissue is higher in the bull than the same tissue in male sheep. In addition, in both animals higher amount of androgens were found in liver in comparison with meat and testis.